Xiaoqing PAN

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SUMMARY

Doctoral candidate in **Microbiology** specializing in RNA biology in the context of host-pathogen interactions. Proficient in R, Linux and NGS data processing within HPC environments (Slurm). Experienced in RNA-seq analysis, prediction of A-to-I RNA editing and alternative splicing and microbial assays. Passionate about integrating bioinformatics to address questions in biotechnology and immunology. Published five papers and two in preparation.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

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PROFESSIONAL EXPERIENCES

Human PBMC transcriptome changes upon fungal stimulations

- Designing a high-throughput **Snakemake workflow** to quantify differential expression and alternative splicing events, characterizing the dynamic response of human PBMCs to fungal pathogens.
- Leading a six people-involved project. Collaborate with Fritz-Lipmann-Institut, Germany
- DOI: 10.1021/acsinfecdis.4c00598

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Develop both sRNA- and mRNA-seq pipeline for differential expression gene analysis.
- Experimental validation support for determining **the role of RNAi machinery and sRNA** using Northern Blot, PCR, qPCR assays.
- Leading and involving a twenty people project. Collaborate with The University of Manchester, the University of Jena
- DOI: 10.1261/rna.079350.122 and DOI: 10.1101/2024.05.24.595671

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analysis of human extracellular small RNA transcriptome.
- Involved in a ten people project. Collaborate with the University of Jena.

Non-coding RNA analysis of rice-*Xoo* interaction

- Experimental validation support for determining **non-coding RNAs (sRNAs, lncRNAs, circRNAs) in rice responding to fungal infections** using PCR, qPCR, LC-MS, assays.
- Leading a five people-involved project. Collaborate with Sun Yat-Sen University, China
- DOI: 10.1186/s42483-023-00188-8 and DOI: 10.1016/j.jplph.2022.153905

Teaching

- Microbiology and Molecular Biology - MMB005, 2022 - 2025

TECHNICAL SKILLS

- **Bioinformatics**: HPC, Snakemake, Genomics, RNA-seq, SHAPE-seq, Alternative splicing, Data visualization

- **Molecular Biology**: Genetic Engineering, Northern Blot, qPCR, PCR , LC-MS, Gibson Assembly Cloning, Cultivation of microorganisms

- **Programming & Tools**: R, python, Bash, Git, Docker, Linux

- **Language:** English (fluent), German (B1), Chinese (native)

PUBLICATIONS

Pan X*, Kelani AA*, *et al*. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microl.

Pan X*, *et al*. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.

Xia K*, **Pan X***, *et al*. (2023). *XOO*-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.

Xia K*, **Pan X***, *et al*. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3, OsNPF2.4,* and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.

Kelani AA, ..., **Pan X**, ..., Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

2025 - Oral presentation: German speaking Mycological Society (DMykG), Germany

2025 - Poster: Gordon Research Seminar - RNA Editing, Italy

2024 - Oral presentation: Jena RNA Club Symposium, Germany

2024 - Poster: RNA Society Meeting, UK // Poster

2023 - Poster: EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- **DAAD Scholarship** 2021 - 2025
- **Jena School for Microbial Communication (JSMC)** travel grant 2023